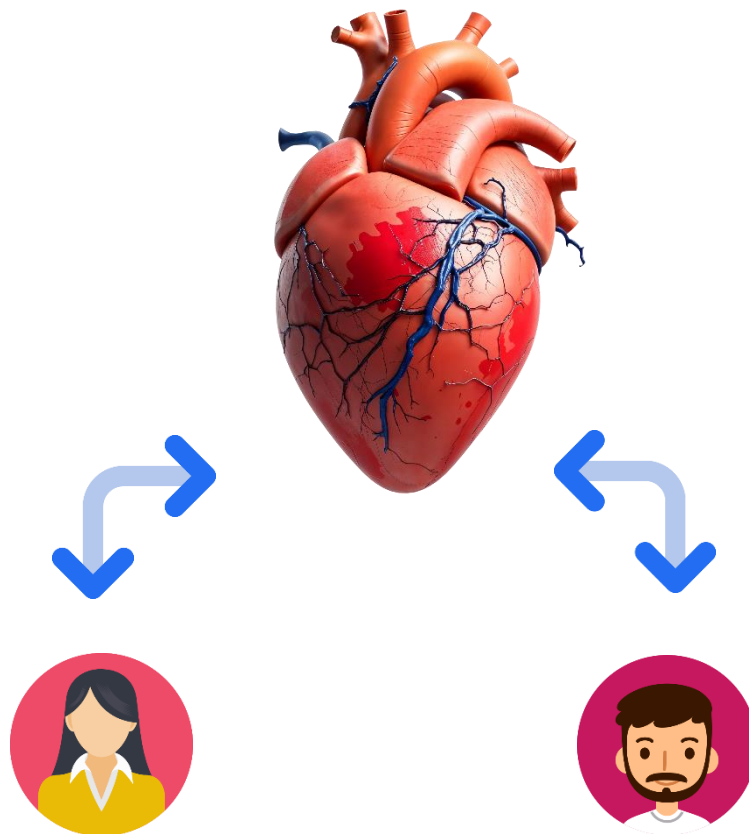


Heart Failure Interactive Dashboard – Case Study Summary



About Dataset:

13 Columns (clinical features): -

- ✓ age: age of the patient (years)
- ✓ anaemia: decrease of red blood cells or Hemoglobin (Boolean)
- ✓ creatinine phosphokinase (CPK): level of the CPK enzyme in the blood (mcg/L)
- ✓ diabetes: if the patient has diabetes (Boolean)
- ✓ ejection fraction: percentage of blood leaving the heart at each contraction (percentage)
- ✓ high blood pressure: if the patient has hypertension (Boolean)
- ✓ platelets: platelets in the blood (Kiloplatelets /mL) - sex: woman or man (binary)
- ✓ serum creatinine: level of serum creatinine in the blood (mg/dL)
- ✓ serum sodium: level of serum sodium in the blood (mEq/L) –
- ✓ smoking: if the patient smokes or not (Boolean)
- ✓ time: follow-up period (days)
- ✓ [target] death event: if the patient died during the follow-up period (Boolean)

Project Overview:

Developed an interactive Tableau dashboard using heart failure clinical records. The dashboard features donut charts, scatter plots, histograms, and box plots to visualize patient demographics, clinical measurements, and outcomes.

Insights Gained:

- Identified key risk factors such as age and serum creatinine levels that are linked to higher mortality risk.
- Observed differences in outcomes based on gender and other patient characteristics.
- Detected outliers and distribution patterns in clinical variables, helping to highlight patients who may require special attention.

Real-World Applications:

- Enables healthcare providers to quickly assess patient risk profiles and tailor treatment plans.
- Supports hospitals in allocating resources effectively to high-risk patient groups.
- Facilitates patient education by presenting complex data in an understandable visual format.
- Assists researchers and policymakers in identifying trends to improve clinical guidelines and public health strategies.

This project demonstrates how interactive data visualization can empower stakeholders to explore complex clinical data and make informed decisions to improve patient care.

1. What problem were you trying to solve?

The goal was to analyse clinical data of heart failure patients to identify patterns and risk factors that affect patient outcomes. By creating an interactive dashboard, the aim was to help healthcare professionals explore the data easily and make informed decisions to improve patient care.

2. What insights did you gain, and how could they be used in real-world scenarios?

- Certain clinical factors like age and serum creatinine strongly correlate with higher risk of heart failure.
- Gender differences in outcomes were observed, suggesting tailored treatment approaches.
- Identifying outliers helped flag patients who may need special attention.
- The interactive dashboard allows clinicians to quickly explore patient data, supporting faster and more personalized care decisions.
- Hospitals can use these insights to allocate resources better and improve patient monitoring strategies.